Sahil Ahamed Shaik

Robotics and AI Engineer | ROS 2 | Autonomous Navigation and Manipulation | AI and Machine Learning

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PROFESSIONAL SUMMARY

Robotics and AI Engineer with experience in ROS 1/2, autonomous navigation, mobile manipulation, robotic perception, and real-world system integration. Proficient in Python and C++ with hands-on experience developing and integrating robotics software across perception, planning, control, simulation, and hardware platforms. Experienced with OpenCV, machine learning, deep learning, and reinforcement learning, with exposure to NVIDIA Isaac Sim and Isaac Lab. Master of Engineering Science in Robotics from UNSW Sydney, with published research and practical experience in autonomous mobile robots, collaborative manipulators, and intelligent robotic systems.

TECHNICAL SKILLS

Robotics and Autonomous Systems: ROS 1/2, Autonomous Navigation, Mobile Manipulation, Motion Planning, Robot Integration

Perception and Estimation: OpenCV, 3D Perception, Pose Estimation

Machine Learning and AI: PyTorch, TensorFlow, Deep Learning, Reinforcement Learning

Simulation: Gazebo, NVIDIA Isaac Sim, NVIDIA Isaac Lab, Webots, CoppeliaSim

Programming and Development: Python, C++, C, MATLAB, Linux, Git, CMake, Visual Studio Code

Robotics Hardware and Platforms: Mobile Robots, Collaborative Manipulators, UGV Platforms, NVIDIA Jetson, Perception Sensors (LiDAR, RGB-D), Arduino

Engineering Tools: Simulink, SolidWorks, ANSYS, AutoCAD, TMflow, Mobile Planner

PROFESSIONAL EXPERIENCE

Master's Thesis Researcher - UNSW, Sydney, Australia Sep 2025 - Jun 2026

ROS 2 Integration and Control of the Omron MoMa Mobile Manipulator

- Developed and integrated a ROS 2 based mobile manipulation framework combining the Omron LD-250 autonomous mobile robot and Techman TM12-S collaborative manipulator to address an autonomous manipulation task for the WARA Robotics Challenge.
- Developed a robotic perception pipeline using ArUco markers and an Intel RealSense RGB-D camera for target detection, depth based 3D localisation, and grasp pose estimation.
- Transformed detected target poses from the camera coordinate frame into the manipulator reference frame to support MoveIt 2 motion planning and grasp execution.
- Developed ARCL based communication interfaces to convert proprietary LD-250 data into ROS 2 compatible odometry, laser scan, point cloud, and robot state information.
- Configured and integrated Nav2, SLAM Toolbox, AMCL, TF2, RViz, URDF/Xacro, and local and global costmaps for mapping, localisation, autonomous navigation, coordinate frame management, and system visualisation.
- Integrated MoveIt 2 and TMflow for manipulator planning and hardware control, coordinating navigation, perception, arm motion, and gripper operation across Gazebo simulation and physical robot hardware.
- Demonstrated the complete perceive, grasp, and navigate sequence end-to-end in Gazebo simulation, and validated arm and gripper control on the physical TM12-S hardware.

ROS Intern - Edgeforce Solutions Dec 2023 - May 2024

- Contributed to robotics projects involving unmanned ground vehicles (UGVs), autonomous navigation systems, airport tug concepts, and defence related robotic platforms.
- Developed and integrated ROS 1/2 based robotic systems, including sensor integration, URDF modelling, navigation, and simulation workflows.
- Worked on path planning and autonomous navigation using Move Base, Dynamic Window Approach (DWA), waypoint navigation, costmaps, SLAM, and custom A* planning.
- Integrated camera and 3D sensing data for obstacle perception and environmental awareness, and designed UGV and condition based monitoring (CBM) components using CAD.

Robotics Intern - Abhyaz Robotics Jun 2022 - Jul 2022

- Worked on robotics development using ROS, Python, Linux, embedded systems, CAD modelling, and TurtleBot teleoperation control.
- Developed CAD and embedded system prototypes using AutoCAD, Tinkercad, and Arduino, with additional exposure to UAV systems.

EDUCATION

Master of Engineering Science (Robotics) 2024 - 2026

University of New South Wales (UNSW Sydney), Sydney, Australia | International Scientia Coursework Scholarship Recipient

Bachelor of Technology in Mechatronics Engineering 2020 - 2024

Hindustan Institute of Technology and Science, Chennai, India | Ranked Second in Class

PROJECTS

Knee Rehabilitation Robot with CNN Based Detection Aug 2022

- Designed six mechanical concepts for a knee rehabilitation robot and developed physical prototypes for two selected designs, focusing on portability, controlled rehabilitation motion, and reduced mechanical complexity.
- Developed a proof-of-concept CNN based knee image classification model using TensorFlow and Keras, evaluating its feasibility for abnormality detection and identifying dataset size and image quality as the key limitations for clinical-grade accuracy.
- Evaluated the proposed mechanisms through CAD modelling, prototyping, and comparative design analysis, with the work contributing to subsequent peer-reviewed research.

Borewell Rescue Robot Jan 2024

- Completed a freelance robotics development project for a borewell rescue system incorporating a soft gripping mechanism, sensing, embedded control, and remote operation.
- Designed and fabricated the robotic system with approximately 80% of mechanical components produced using 3D printing and readily available off-the-shelf components.
- Designed and validated mechanical components using SolidWorks and ANSYS, with Arduino based control implemented for real-time operation.

Homogeneous Swarm Robotics in a Virtual Environment Jan 2022

- Developed a multi-robot leader-follower coordination system using TurtleBot3 platforms in Webots, with Python based inter-robot communication and centralized coordination.

PUBLICATIONS AND PRESENTATIONS

- Design and Development of Knee Rehabilitation Robot. IAES International Journal of Robotics and Automation (IJRA), Vol. 13, No. 4, pp. 423-431, Dec 2024. DOI: 10.11591/ijra.v13i4.pp423-431.
- Design and Evaluation of Various Knee Assistive Mechanisms for Rehabilitation. AIP Conference Proceedings, Vol. 2946, 020003, 2023. DOI: 10.1063/5.0179010.
- Presented research work at RIACT 2022, RANE 2023, and AIIMS-3.0; patent application in progress for the knee rehabilitation robot design.

AWARDS AND TRAINING

- International Scientia Coursework Scholarship, UNSW Sydney; ranked second in class in B.Tech Mechatronics Engineering.
- First place in paper presentations at multiple universities, including IIT Madras, VIT, and MIT Chennai.
- Training: TensorFlow 2.0, Introduction to AI and Machine Learning with Python, MATLAB and Simulink Onramp, ROS for Beginners, SMC Pneumatics.